

Session 08

Semantic HTML5, Forms, Tables & Media

What are you going to learn in this session?

In this session, you will move from basic HTML usage to building more meaningful and structured web pages using semantic HTML. You will also learn how to present structured data using tables, collect user input using forms, and enrich your pages with media like images and videos.

By the end of the session, you will build a small “Online Bookstore” page that combines semantic structure, a data table, a user form, and embedded media. The goal is to make you comfortable with real-world page structure rather than memorizing tags.

Why is this topic important?

As web pages grow, using only generic tags like makes code hard to read, maintain, and scale. Semantic HTML introduces tags that carry meaning, helping both humans and machines understand your content better.

Good structure improves readability for developers, accessibility for users (including screen readers), and discoverability for search engines. These are foundational practices you will carry into every project you build.

How should you prepare yourself to attend this session?

Come prepared to code along in VS Code with Live Server running so you can instantly see changes in the browser. You should be comfortable creating files and navigating folders, as well as writing basic HTML from the previous session.

Keep a mindset of “build first, refine later.” Focus on understanding structure and intent rather than perfect syntax. Small mistakes are expected and part of the learning process.

What should you do after the session?

After the session, extend the bookstore page by adding more rows in the table, additional form fields, and more sections using semantic tags. Try replacing generic containers with meaningful tags wherever possible.

Revisit your previous portfolio and refactor it using semantic elements. This practice will strengthen your understanding of structure and improve the quality of your code.

Class Notes

Tables: Displaying Structured Data

Tables are used to display structured data in rows and columns, such as lists of books, users, or transactions. They should not be used for page layout, but only when the data is naturally tabular.

A basic table includes a header section for column titles and a body section for the actual data. This grouping improves readability and makes the structure clear.

```
<table>

  <thead>

    <tr>

      <th>Title</th>

      <th>Author</th>

      <th>Genre</th>

      <th>Price</th>

    </tr>

  </thead>

  <tbody>

    <tr>

      <td>Atomic Habits</td>

      <td>James Clear</td>
```

```
<td>Self-help</td>

<td>₹499</td>

</tr>

<tr>

<td>Clean Code</td>

<td>Robert C. Martin</td>

<td>Programming</td>

<td>₹699</td>

</tr>

</tbody>

</table>
```

Focus on logical grouping using `thead` and `tbody`, and clear column names using `th`. Styling will be handled later with CSS.

Forms: Collecting User Input

Forms allow users to send data to your application, such as signing up for a newsletter or placing an order. Each form contains inputs that capture specific types of data.

Common inputs include text, email, radio buttons, checkboxes, dropdowns, and a submit button. Attributes like `name`, `type`, `placeholder`, and `required` define how inputs behave.

```
<form>

  <label>Name:</label>

  <input type="text" name="name" placeholder="Enter your name"
required />

  <label>Email:</label>
```

```
<input type="email" name="email" placeholder="Enter your email"
required />
```

```
<label>Preferred Genre:</label>
```

```
<select name="genre">
```

```
  <option>Fiction</option>
```

```
  <option>Self-help</option>
```

```
  <option>Programming</option>
```

```
</select>
```

```
<button type="submit">Subscribe</button>
```

```
</form>
```

The focus here is on structure and capturing input correctly. Validation logic and styling will be covered later.

Semantic HTML5 Tags (Core Focus)

Semantic tags give meaning to different parts of your page, making your code easier to read and understand. Instead of using generic containers everywhere, you describe the purpose of each section.

Common semantic tags include several important elements that you will frequently use while structuring real-world pages. Each of these tags carries meaning, which helps both developers and browsers understand the intent of different sections.

- `<header>` is used for the top section of a page or a section, usually containing titles or branding
- `<nav>` is used specifically for navigation links so users can move between pages
- `<section>` is used to group related content together in a meaningful way
- `<article>` is used for standalone content such as blog posts or independent sections
- `<aside>` is used for side content like tips, ads, or additional information

- `<footer>` is used for the bottom section of a page, often containing copyright or contact details

```
<header>
```

```
  <h1>Online Bookstore</h1>
```

```
</header>
```

```
<nav>
```

```
  <a href="#">Home</a>
```

```
  <a href="#">Books</a>
```

```
  <a href="#">Contact</a>
```

```
</nav>
```

```
<section>
```

```
  <h2>Featured Books</h2>
```

```
</section>
```

```
<footer>
```

```
  <p>© 2026 Bookstore</p>
```

```
</footer>
```

The goal is not perfection but clarity. Use tags where they make sense so anyone reading your code can understand the structure instantly.

Media Elements & Entities

Web pages often include images, videos, or audio to improve user experience. HTML provides simple tags to embed these elements directly into your page.

```

```

```
<video controls width="400">
```

```
  <source src="promo.mp4" type="video/mp4" />
```

```
  Your browser does not support video.
```

```
</video>
```

The alt attribute in images is important for accessibility, as it describes the image content for screen readers and fallback scenarios.

HTML entities are used to display reserved characters that would otherwise be interpreted as code. For example, `<` displays the less-than symbol and `>` displays the greater-than symbol.

```
<p>Use &lt;h1> to create a heading.</p>
```

Class Project Build (Guided)

You will create a project folder named `online_bookstore` and build an `index.html` file that combines all the concepts learned in this session. The page should include a semantic layout, a table of books, a simple form, and embedded media.

```
<link rel="stylesheet" href="styles.css" />
```

You will link a pre-built CSS file to see how styling changes the visual appearance of your page. The focus is not on writing CSS, but on understanding how structure and styling connect.

Common Beginner Confusions

Learners often confuse when to use tables versus regular layout elements. Tables should only be used for structured data, not for designing page layouts.

Another common issue is form inputs not working as expected, which is usually due to missing attributes like `name` or `type`. Media elements may also fail to load if file paths are incorrect.

Semantic tags can feel confusing initially, but the goal is simply to replace generic containers with more meaningful ones wherever possible.

Key Takeaways

You can now structure pages using semantic HTML, making your code more readable and meaningful. You have learned how to display structured data using tables and collect user input using forms.

You can also embed images and videos, and understand how HTML entities work. These concepts together allow you to build more realistic and complete web pages that resemble actual products.

At this stage, focus on usage and repetition. The more you build, the more natural these patterns will become.